10ma Clase 24/10/2025

75.43 Introducción a los Sistemas Distribuidos
95.60 Redes y Aplicaciones Distribuidas
TA048 Redes

Tema: Capa de enlace (II)

Desde *6.4 Switched Local Area Networks* hasta el final del capítulo *6* del libro *Computer Networking : A Top-Down Approach. James Kurose and Keith Ross. Publisher: Pearson Edition: 8th, 2021.*

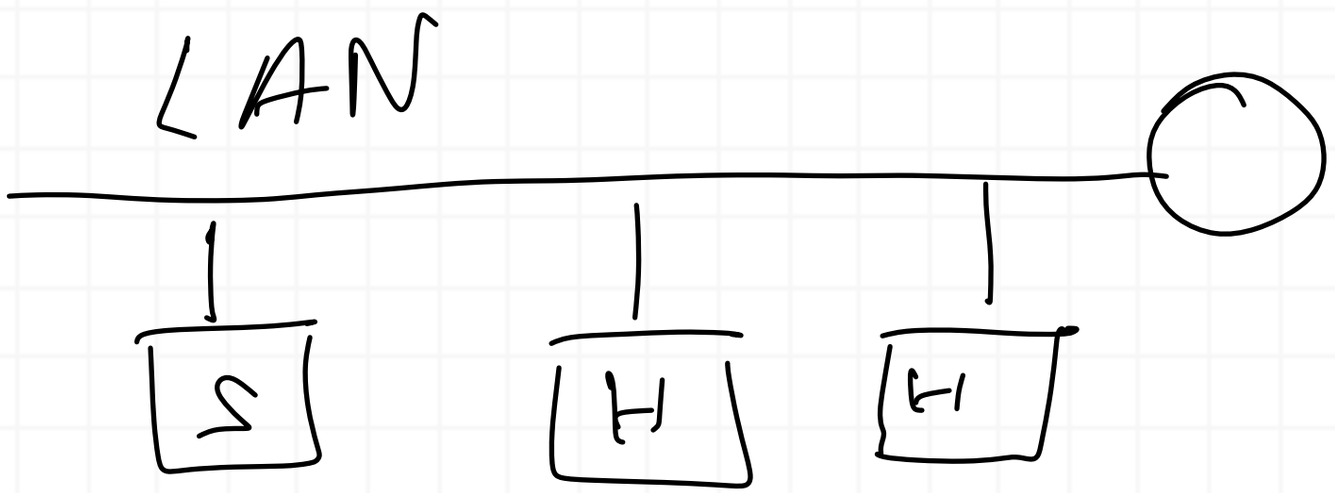
Dr. Ing. J. Ignacio Alvarez-Hamelin

Clase de hoy

- Redes de Área Local (LANs)
- Ethernet
- Otras tecnologías

¿Qué es una LAN (Local Area Netwok)?

- Redes de Area Local
- Direccionamiento
- *Routers vs. Switches*
- ARP (Address Resolution Protocol)



@MAC

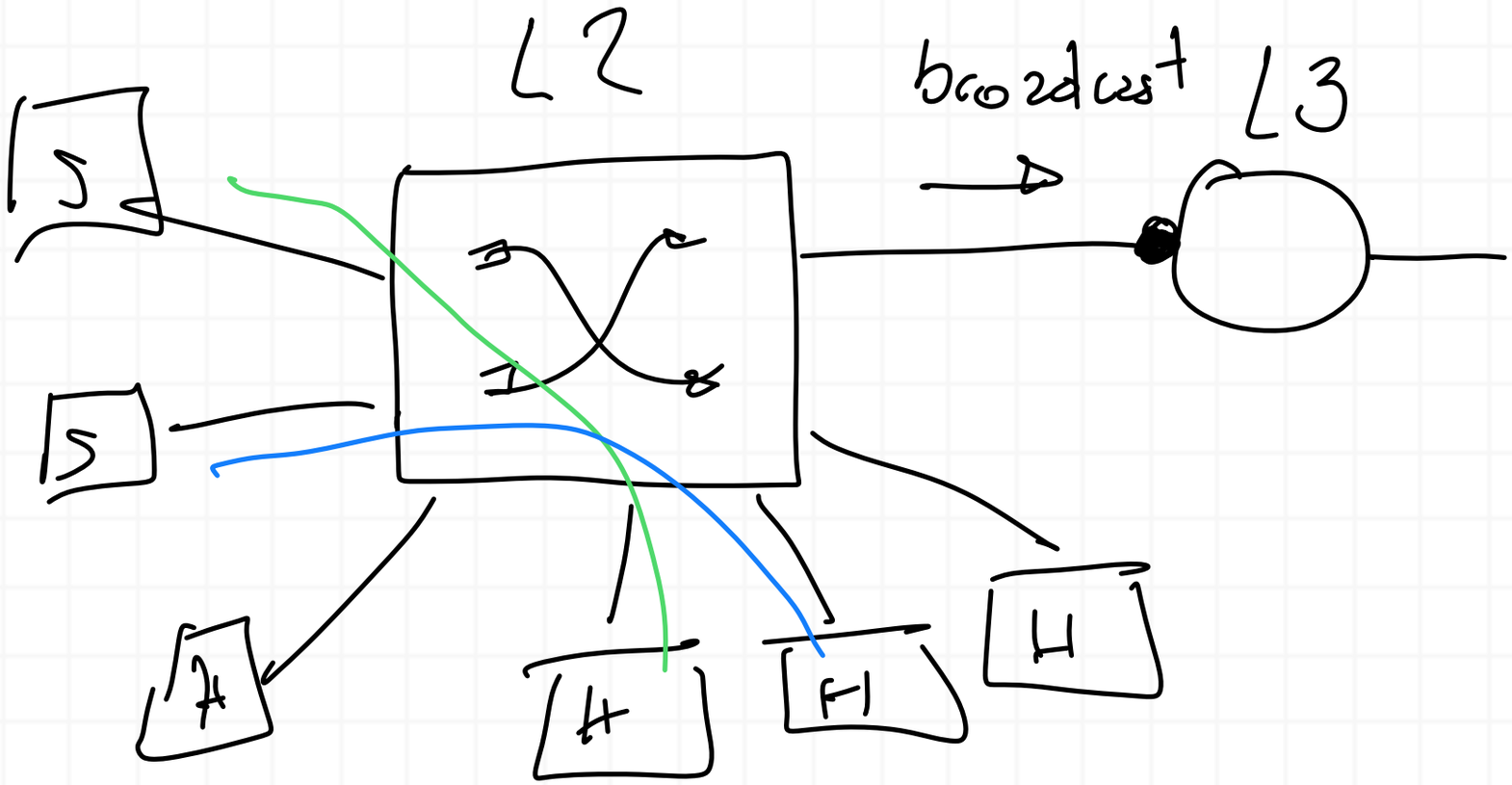
48 bits

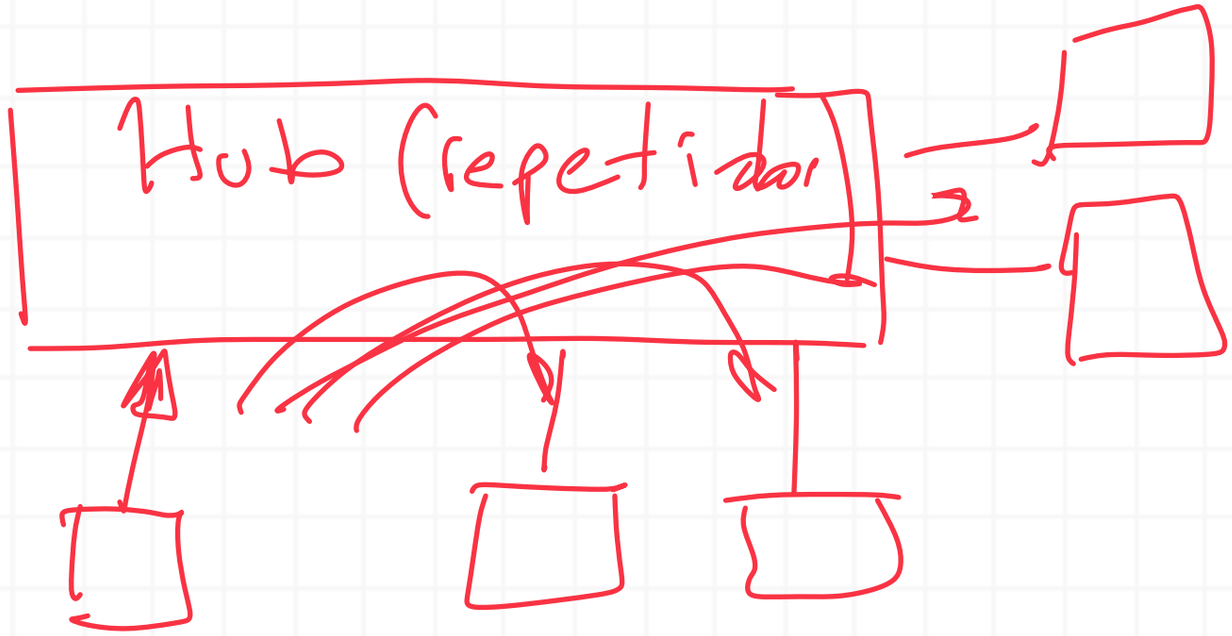
24

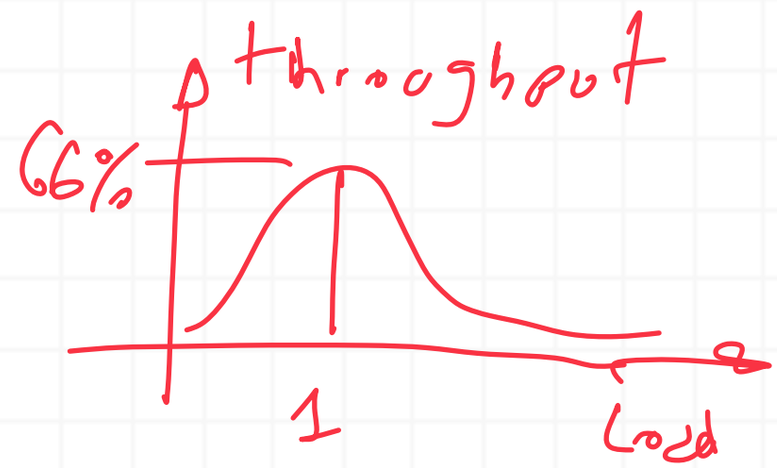
Vendor	

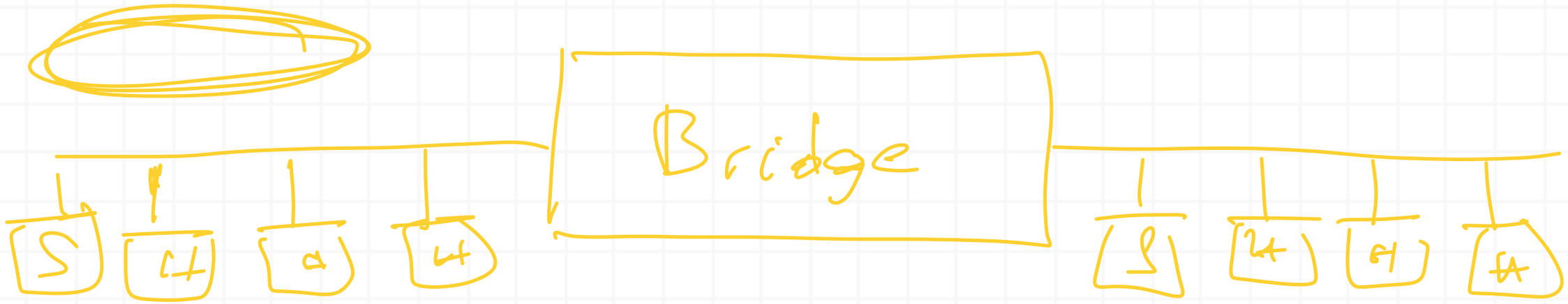
FF:FF:FF:FF:FF:FF

Multicast


L2
broadcast
L3
S
S
H
H
H
H


Hub (repetidor)


throughput
66%
1
Load



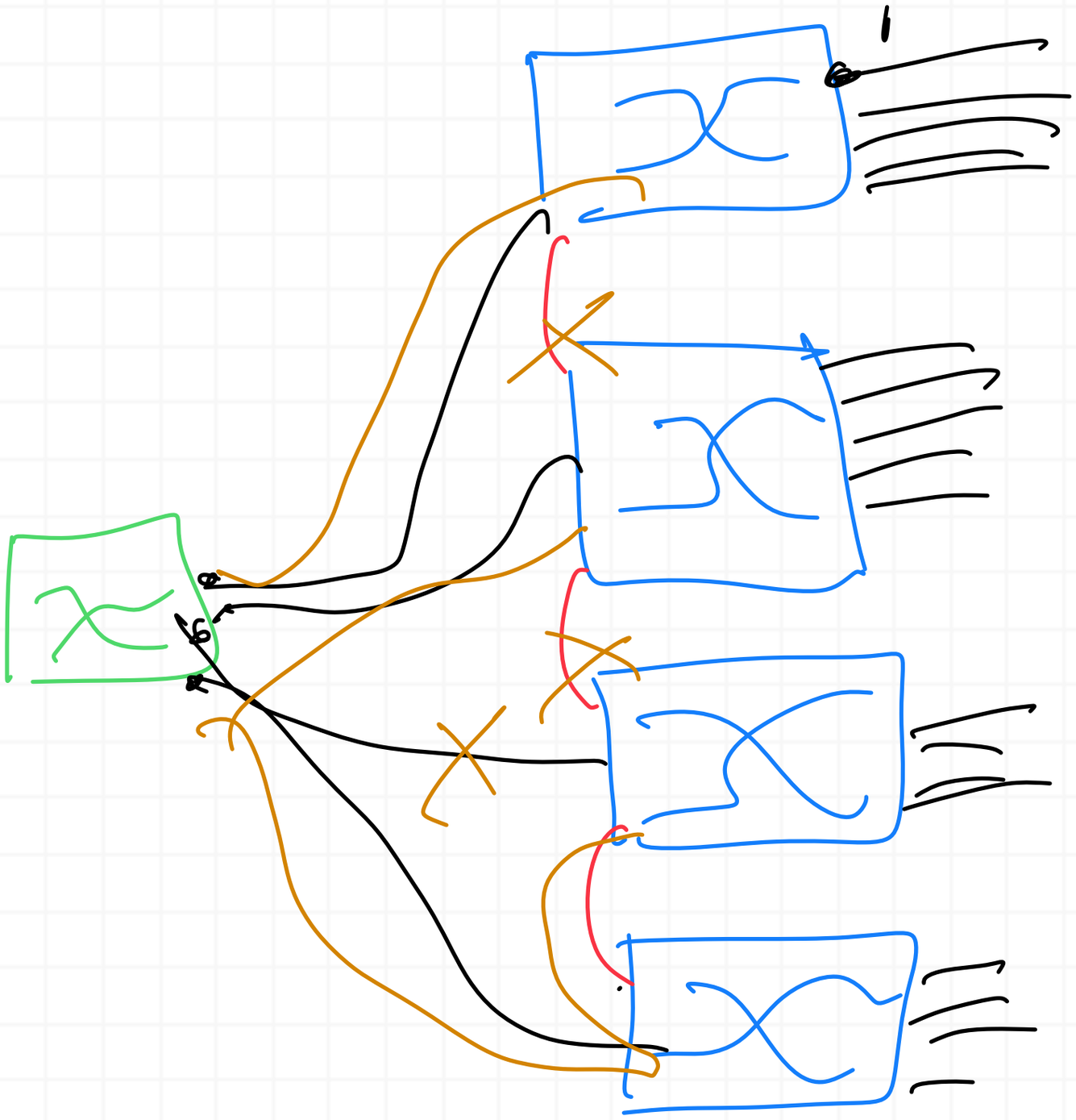
@MAC_destino	@MAC_origen	XXX	IP

L2

A → C

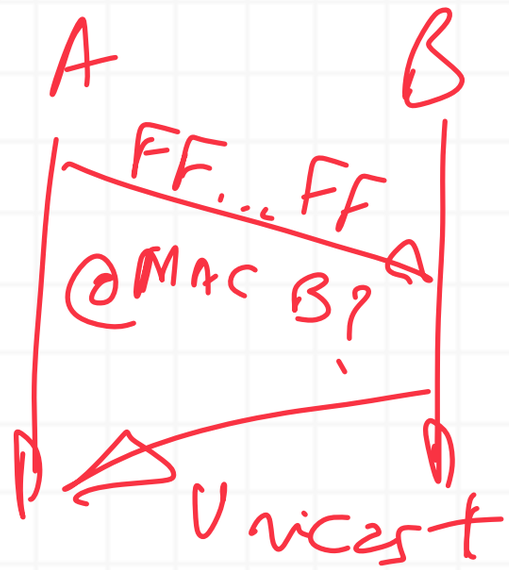
A ← C

Izq.	Der.
A	C
D	B

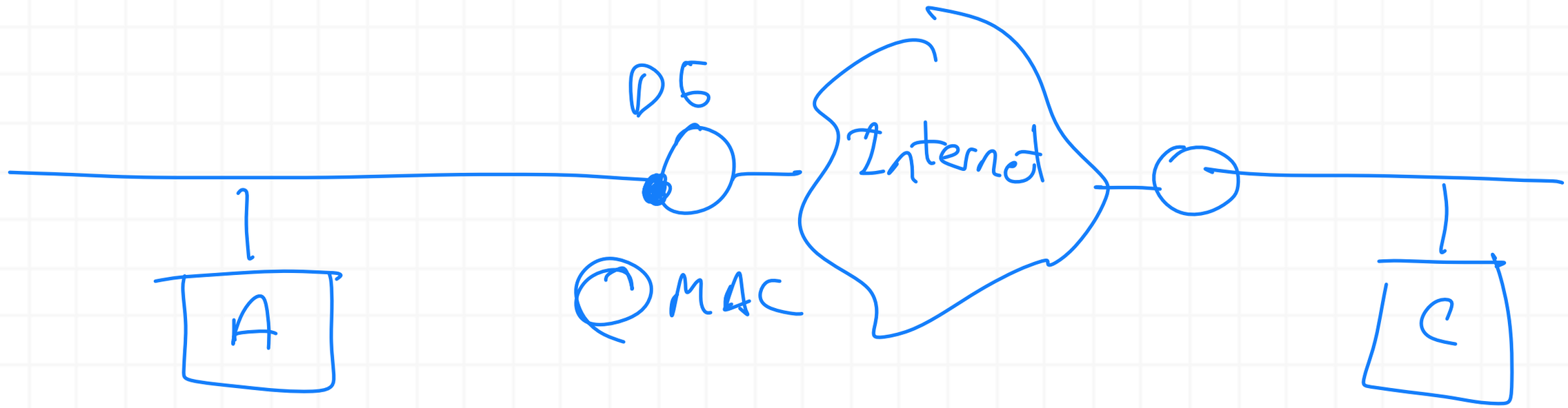


Spanning Tree Protocol

ARP



@MAC_origen | FF...FF | IP_origen | IP_destino

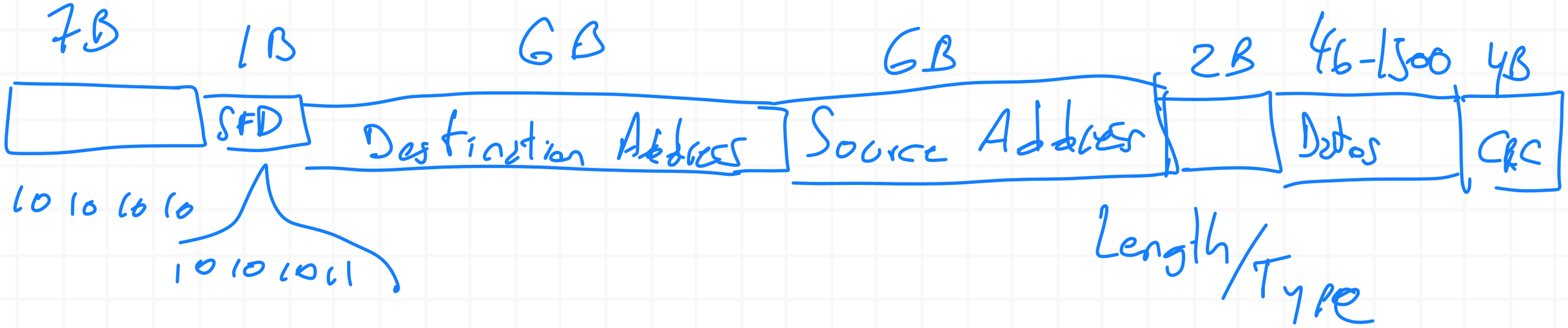


@MAC_origen	@MAC_dest		IP_origen	IP_destino
@MAC_A	@MAC_DG	xx	A	C

Ethernet

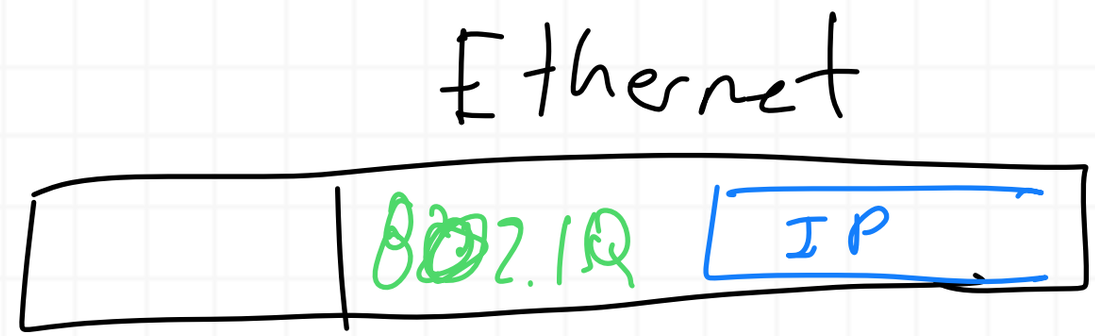
- Ethernet (*Hubs vs. Switches*)
- Estructura del paquete
- *Routers, Switches & SDN*

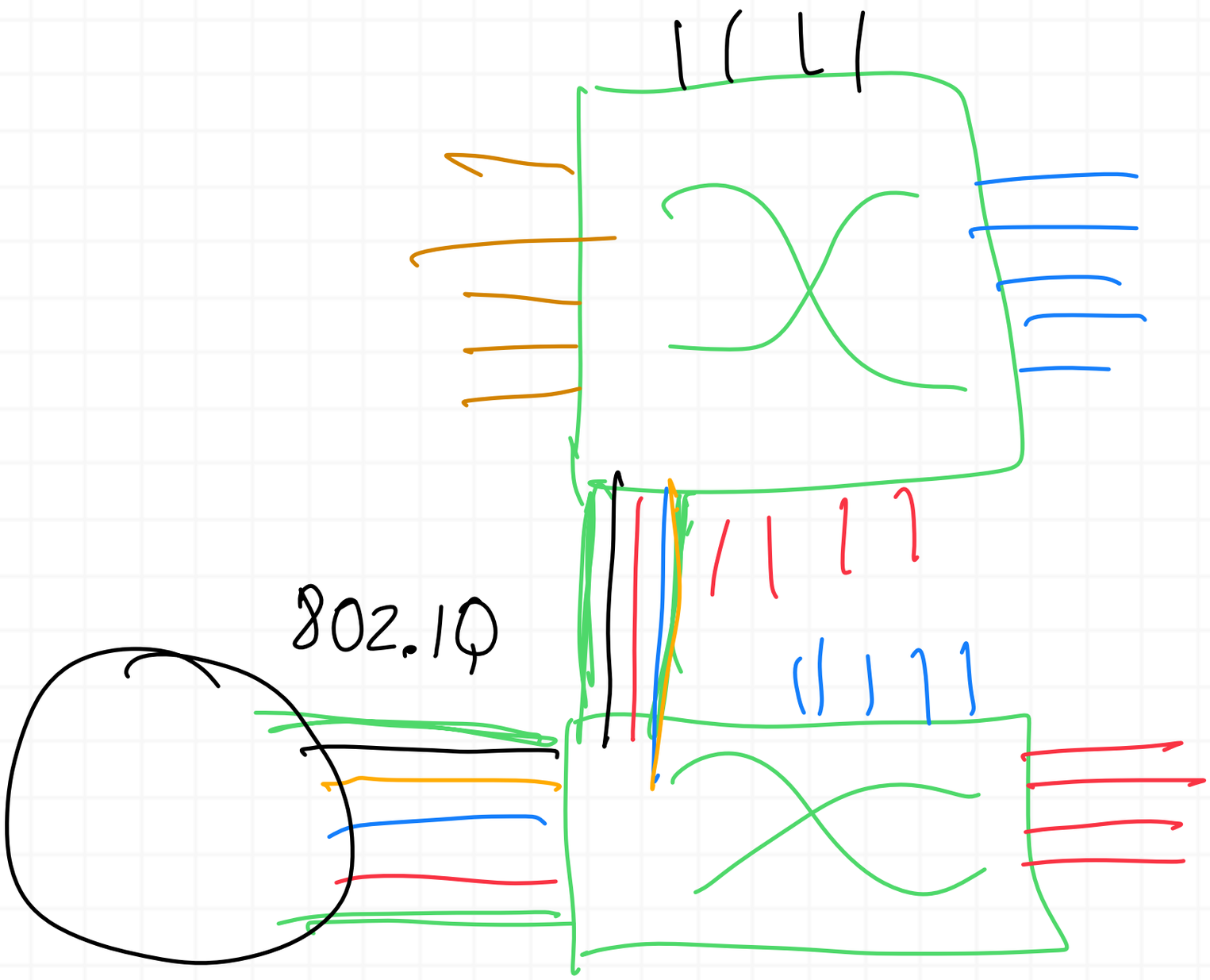
IEEE 802.3 BaseS, Base2, Base T



Otras tecnologías

- VLANs
- Motivación
- IEEE 802.1Q el protocolo del colector (*trunk*)
- MPLS (*Multiple-Path Label Switching*)

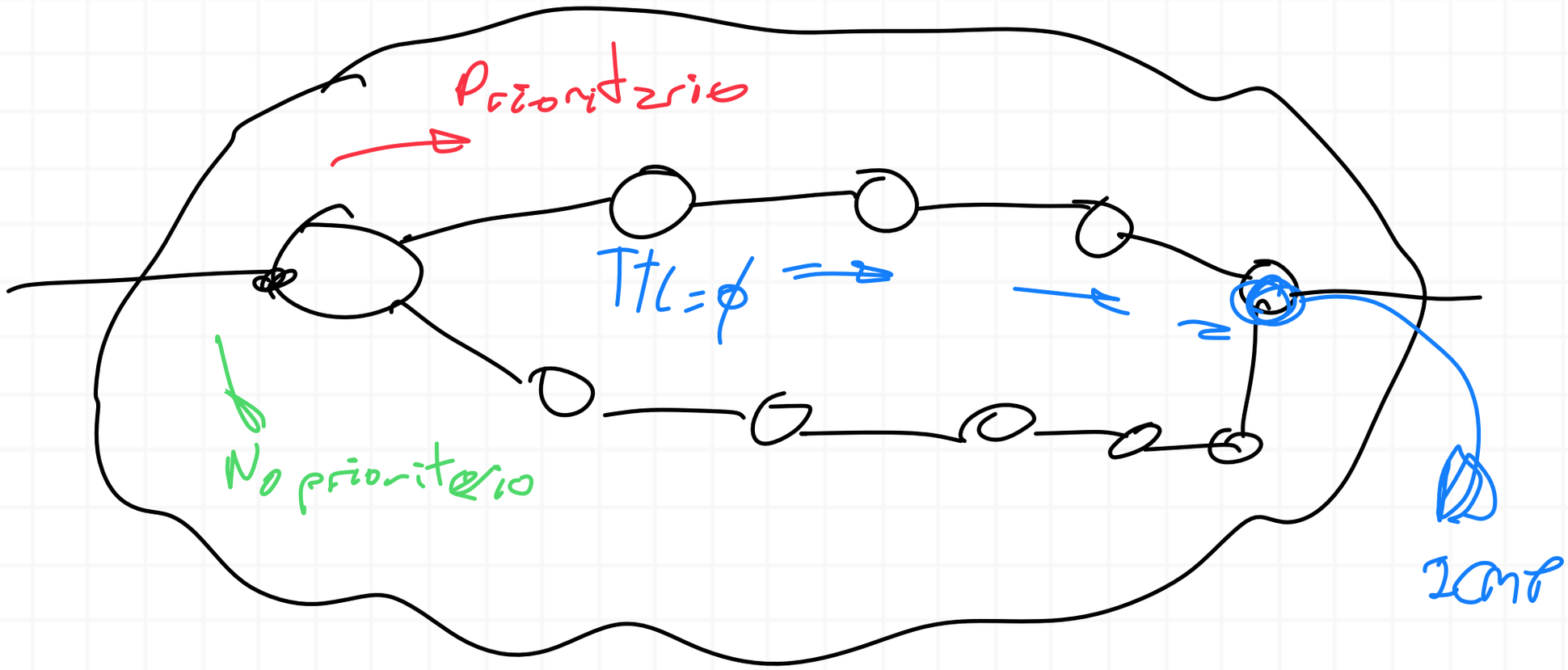




MPLS

MPLS | IP

etiqueta ≡ Flow Label IPv6



¿Cómo se ejecuta en cada una de las capas una solicitud de una página web?

Aplicación : http : DNS $\xrightarrow{\text{recursiva}}$ servidor

transporte : TCP UDP

Red : @IP ? Routing

Enlace : ARP (@MAC destino : Broadcast
@IP destino : next hop
[gateway])

Lectura para la próxima clase:

Leer hasta el Capítulo 7 hasta *7.3 WiFi: 802.11 Wireless LANs,* inclusive.